

Cardiovascular Agents

Cardiac Disorders

- Heart diseases can be primarily grouped into three major disorders:
 - Cardiac failure,
 - Ischemia and
 - Cardiac arrhythmia.

- **Cardiac failure (CHF)**

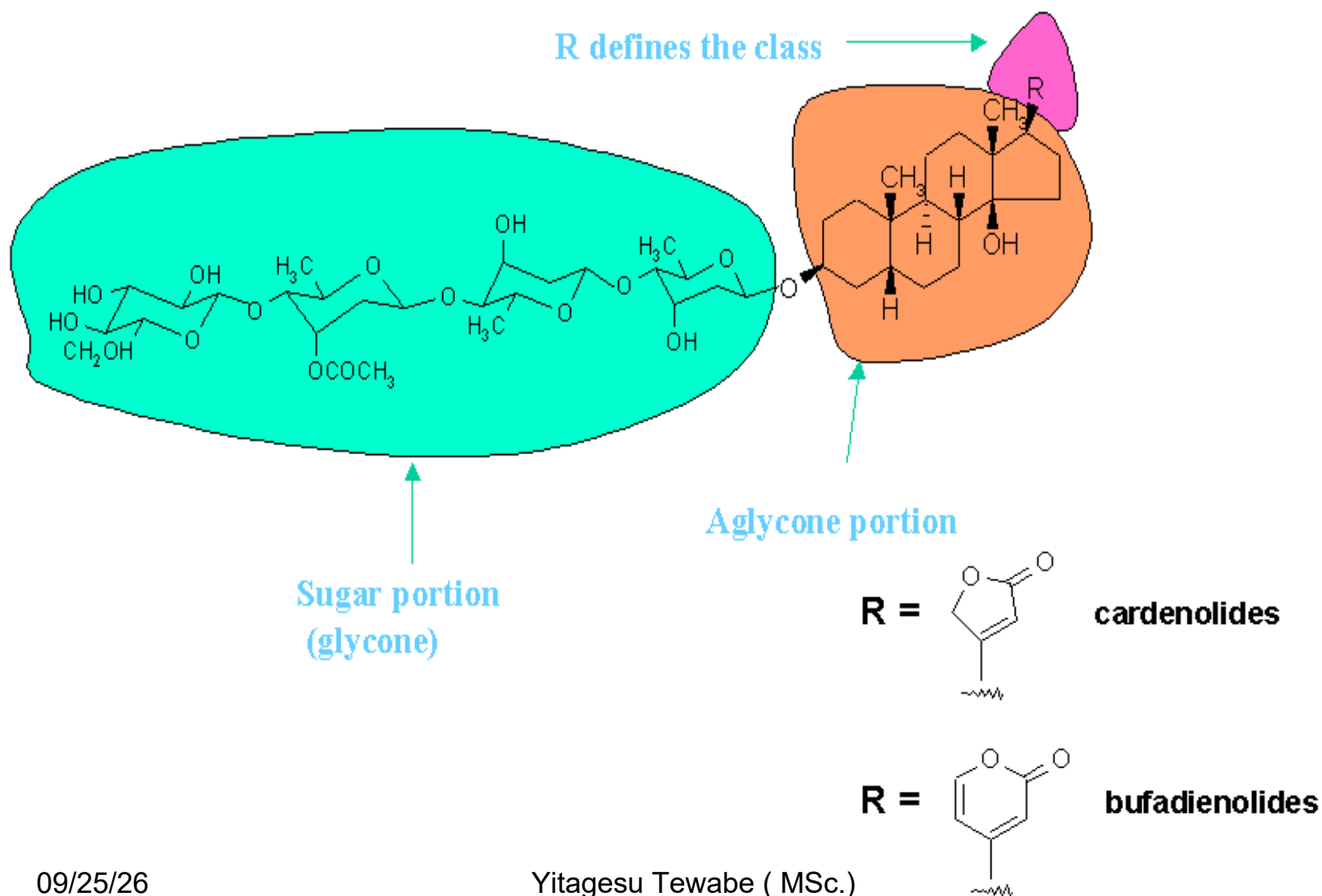
- Inability of the heart to pump blood effectively at a rate that meets the needs of the metabolizing tissues.
- This occurs when the muscles that perform contraction and force the blood out of heart are performing weakly.
- Thus cardiac failures primarily arise from the reduced contractility of heart muscles, especially the ventricles.

- Reduced contraction of heart leads to reduced heart output but new blood keeps coming in resulting in the increase in heart blood volume.
 - The heart feels congested. Hence the term congestive heart failure.
- Congested heart leads to lowered blood pressure and poor renal blood flow.
 - This results in the development of edema in the lower extremities and the lung (pulmonary edema) as well as renal failure.

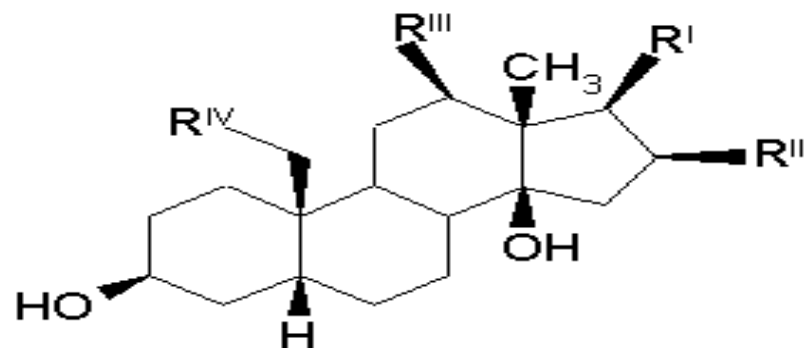
- **Cardiac Glycosides**

- Increasing the force of contraction of the heart (positive inotropic activity) is very important for most heart failure patients.
 - Cardiac steroids are perhaps the most useful drugs that increase heart contractility.
 - The cardiac glycosides are an important class of naturally occurring drugs whose actions include both beneficial and toxic effects on the heart.

- Cardiac steroids are widely used in the modern treatment of CHF and for treatment of atrial fibrillation and flutter.
 - Yet their toxicity remains a serious problem.
- **Structure**
 - Cardiac glycosides are composed of two structural features :
 - The sugar (glycoside) moiety
 - The non-sugar (aglycone - steroid) moieties.

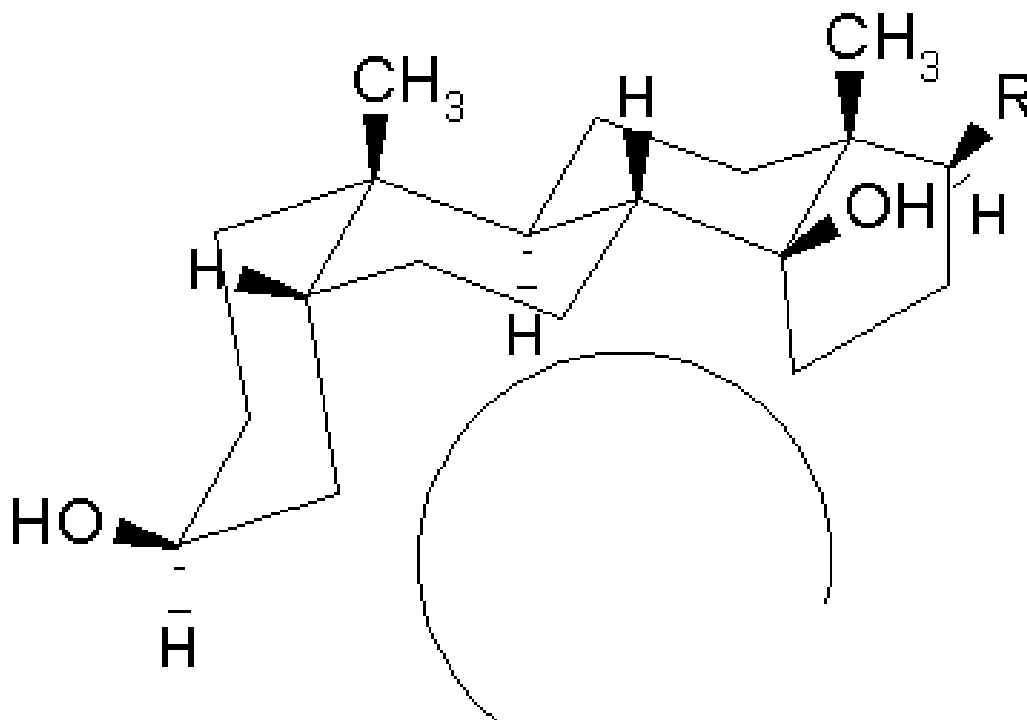


- The R group at the 17-position defines the class of cardiac glycoside.
- Two classes have been observed in Nature –
 - The cardenolides and the bufadienolides.
 - The cardenolides have an unsaturated butyrolactone ring while the bufadienolides have an α -pyrone ring.



	R ^I	R ^{II}	R ^{III}	R ^{IV}
Digitoxigenin		H	H	H
Digoxigenin		H	OH	H
Gitoxigenin		OH	H	H
Ouabagenin		H	H	H
Strophanthidin		H	H	=O
Bufalin		H	H	H

- **The aglycone moiety:**
 - The steroid nucleus has a unique set of fused ring system that makes the aglycone moiety structurally distinct from the other more common steroid ring systems.
 - Rings A/B and C/D are *cis* fused while rings B/C are *trans* fused.
 - Such ring fusion gives the aglycone nucleus of cardiac glycosides the characteristic '**U**' shape.



- The steroid nucleus has hydroxyls at 3- and 14- positions of which the sugar attachment uses the 3-OH group.
- 14-OH is normally unsubstituted.
- Many genins have OH groups at 12- and 16-positions.
 - These additional hydroxyl groups influence the partitioning of the cardiac glycosides into the aqueous media and greatly affect the DOA.

- The lactone moiety at C-17 position is an important structural feature.
- The size and degree of unsaturation varies with the source of the glycoside.
- Normally plant sources provide a 5-membered unsaturated lactone while animal sources give a 6-membered unsaturated lactone.

- **Sugar moiety:**
- One to four sugars are found to be present in most cardiac glycosides attached to the 3b-OH group.
- The sugars most commonly used include:
 - L-rhamnose, D-glucose, D-digitoxose, D-digitalose, D-digginose, D-sarmentose, L-vallarose, and D-fructose.
- These sugars predominantly exist in the cardiac glycosides in the β -conformation.

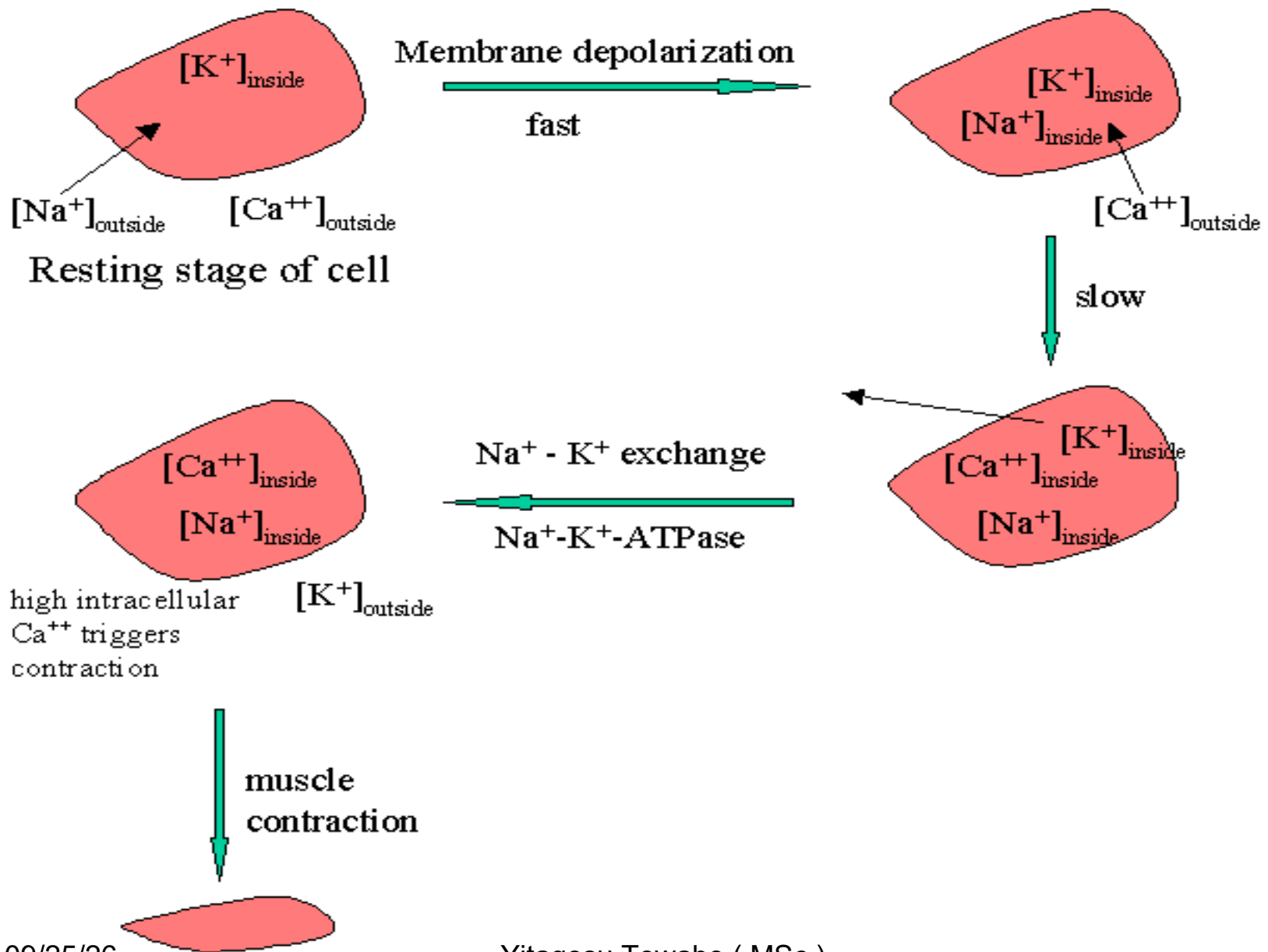
- The presence of acetyl group on the sugar affects
 - Lipophilic character and the kinetics of the entire glycoside.
- Because the order of sugars appears to have little to do with biological activity
 - Nature has synthesized a repertoire of numerous cardiac glycosides with differing sugar skeleton but relatively few aglycone structures.

- **Structure - Activity Relationships**
 - The sugar moiety appears to be important only for the partitioning and kinetics of action
 - It possesses no biological activity.
 - For example, elimination of the aglycone moiety eliminates the activity of alleviating symptoms associated with cardiac failure.

- The "backbone" U shape of the steroid nucleus appears to be very important.
- Structures with C/D *trans* fusion are inactive.
- Conversion to A/B *trans* system leads to a marked drop in activity.
 - Thus, although not mandatory A/B *cis* fusion is important.
- The 14 β -OH groups are now believed to be dispensable.
 - A skeleton without 14 β -OH group but retaining the C/D *cis* ring fusion was found to retain activity.

- Lactones alone, when not attached to the steroid skeleton, are not active.
 - Thus the activity rests in the steroid skeleton.
- The unsaturated 17-lactone plays an important role in receptor binding.
 - Saturation of the lactone ring dramatically reduced the biological activity.
- The lactone ring is not absolutely required.
 - For example, using α,β -unsaturated nitrile (C=C-CN group), the lactone could be replaced with little or no loss in biological activity.

- **Biochemical Mechanism of Action**
 - The mechanism whereby cardiac glycosides cause a positive inotropic effect and electrophysiologic changes is still not completely clear.
 - Several mechanisms have been proposed, but the most widely accepted involves
 - The ability of cardiac glycosides to inhibit the membrane bound $\text{Na}^+\text{-K}^+\text{-ATPase}$ pump responsible for $\text{Na}^+\text{-K}^+$ exchange.



- The process of membrane depolarization / repolarization is controlled by the movement of three cations
 - Na^+ , Ca^{+2} , and K^+ , in and out of the cell.
- At the resting stage, the concentration of Na^+ is high on the outside.
- On membrane depolarization sodium fluxes-in leading to an immediate elevation of the action potential.

- Elevated intracellular Na^+ triggers the influx of free Ca^{++} that occurs more slowly.
 - The higher intracellular $[\text{Ca}^{++}]$ results in the efflux of K^+ .
- The re-establishment of the action potential occurs later by the reverse of the Na^+ - K^+ exchange.
- The Na^+ / K^+ exchange requires energy which is provided by an enzyme Na^+ - K^+ -ATPase.

- Cardiac glycosides are proposed to inhibit this enzyme with a net result of reduced sodium exchange with potassium that leaves increased intracellular Na^+ .
 - This results in increased intracellular $[\text{Ca}^{++}]$.
- Elevated intracellular calcium concentration triggers a series of intracellular biochemical events that ultimately result in an increase in the force of the myocardial contraction or a positive inotropic effect.

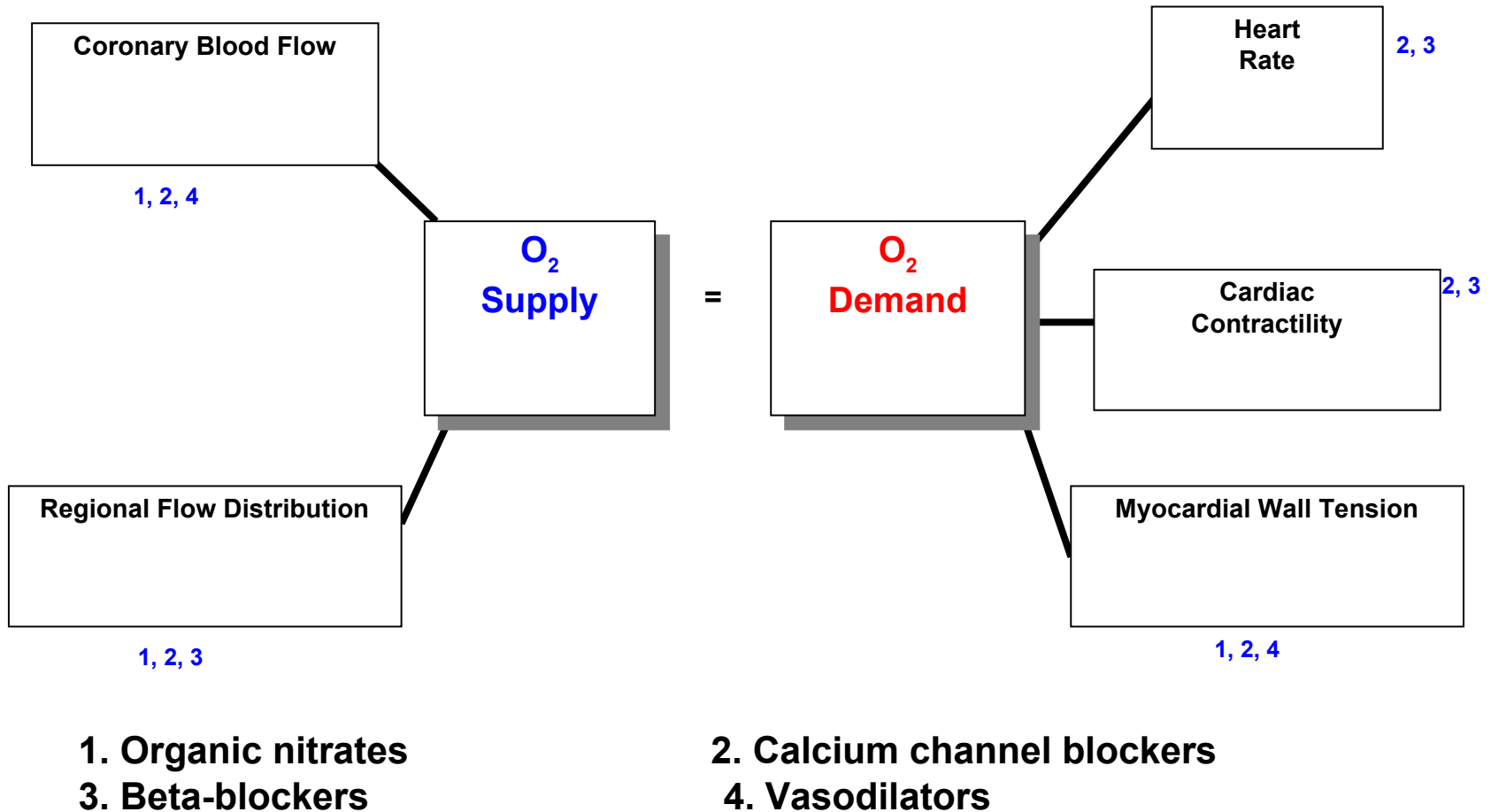
Antianginal agents & vasodilators

- **Angina pectoris:** is the principle symptom of ischemic heart disease
 - Caused by an imbalance between myocardial oxygen demand and oxygen supply by coronary vessels.
 - Such an imbalance can result from increased myocardial oxygen demand caused by exercise, decreased myocardial oxygen delivery, or both.

- Angina pectoris is always associated with sudden, severe chest pain and discomfort, although some individuals do not experience pain with ischemia.
- Thus, angina occurs because the blood supply to the myocardium through coronary vessels is insufficient to meet the metabolic needs of the heart muscle for oxygen

- **Goal of Antianginal Drugs:** to restore balance between Myocardial O₂ Demand and Supply

Classes and MOA of Anti-anginal Drugs

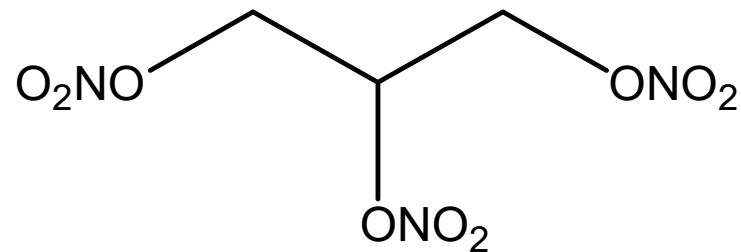


Organic nitrates

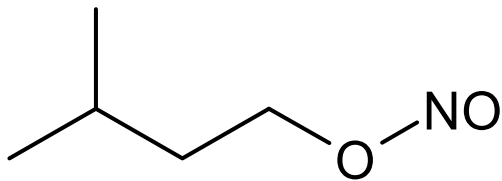
- Nitrovasodilators were used in prophylaxis & treatment of angina
 - Their importance reduced due to the introduction of
 - **Calcium channel blockers**
 - **∇ β -blockers**
- But are still of choice for acute anginal episodes

- Nitroglycerin

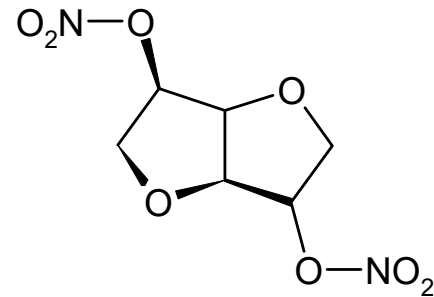
- One of the first & still important member of the group



Glyceryl trinitrate



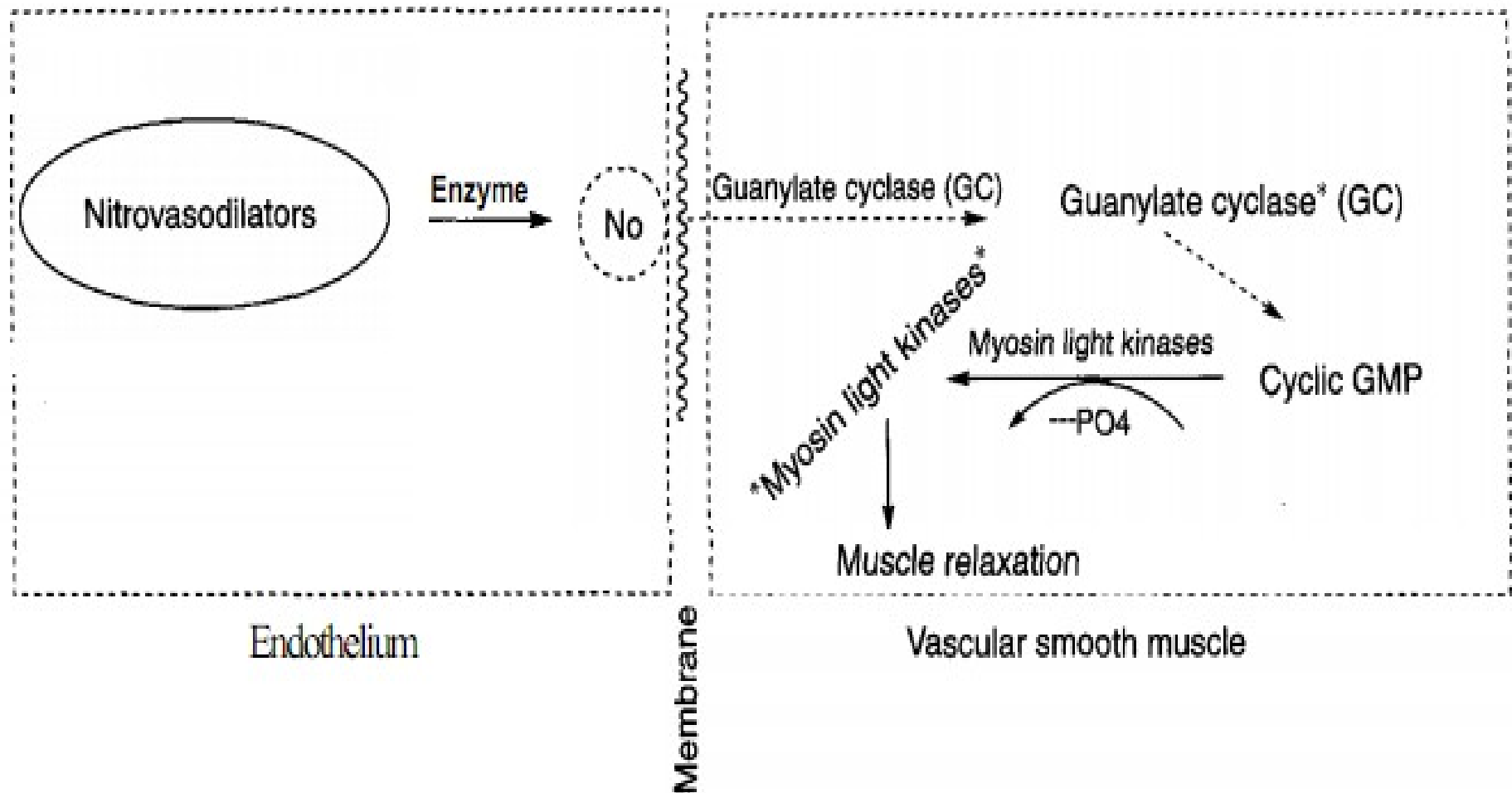
Isopentyl Nitrite



Isosorbid dinitrate

- Nitrovasodilators
 - Cause relaxation of vascular smooth muscle by generating NO *in situ*

MOA of nitrovasodilators



Calcium channel blockers

- Introduced into clinical medicine in 1960's
- Calcium is vital to
 - Generation of nerve impulses
 - Heart muscle contraction
- Stimulation of heart cell causes ion channel activation
 - Ca enters cell stimulating further Ca release from within the cell
 - This in turn initiates contraction

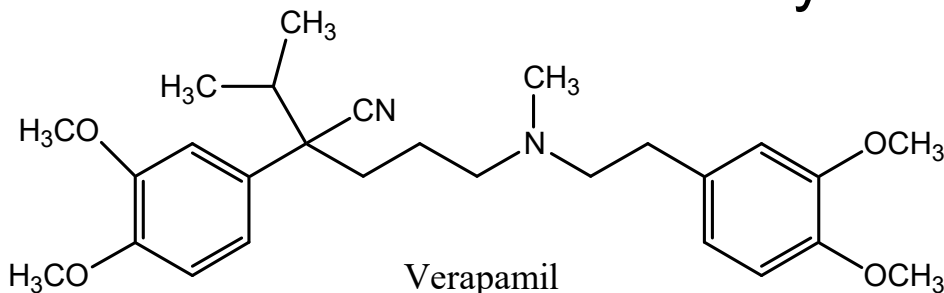
- All Ca antagonists are vasodilators
 - Basis for use in
 - Angina
 - Hypertension
- Inhibition of Ca influx into cardiac tissue
 - Basis for their use as antiarrhythmic agents

- **Three classes of Calcium Channels Blockers (CCBs)**
 - Phenylalkylamines
 - Benzothiazepines
 - Dihydropyridines

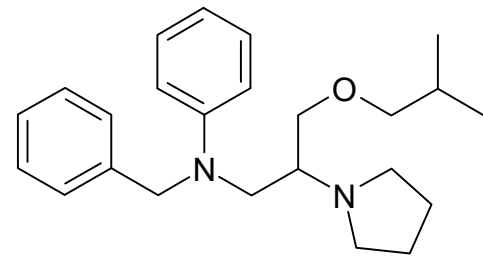
Phenylalkylamines

- **Verapamil**

- A coronary vasodilator with Ca channel blocking activity
- (-) form more potent
- Rapidly absorbed after oral administration
- Quickly metabolized in liver
 - Has low bioavailability



Verapamil

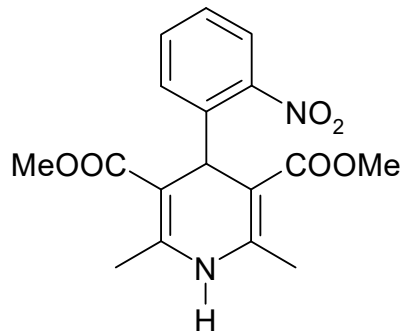


Bepridil

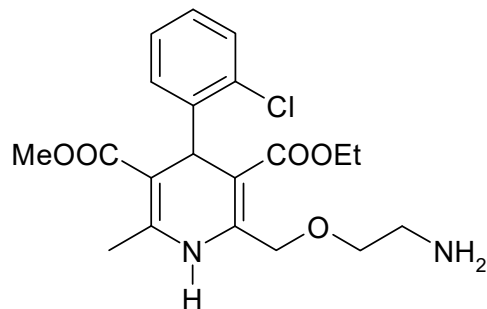
Dihydropyridines

- Nifedipine
 - Most potent of the group as a vasodilator
 - Used to treat hypertension as well as angina
 - Members have greater vascular selectivity
 - Less cardiac depression, for a given degree of vasodilation

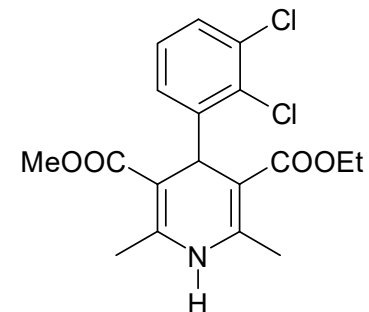
Dihydropyridines...



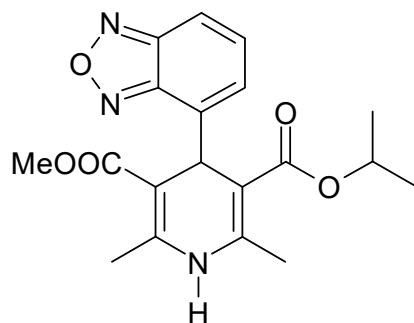
Nifedipine



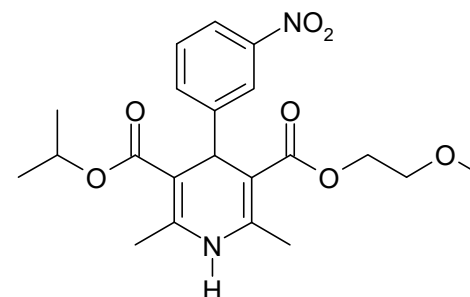
Amlodipine



Felodipin



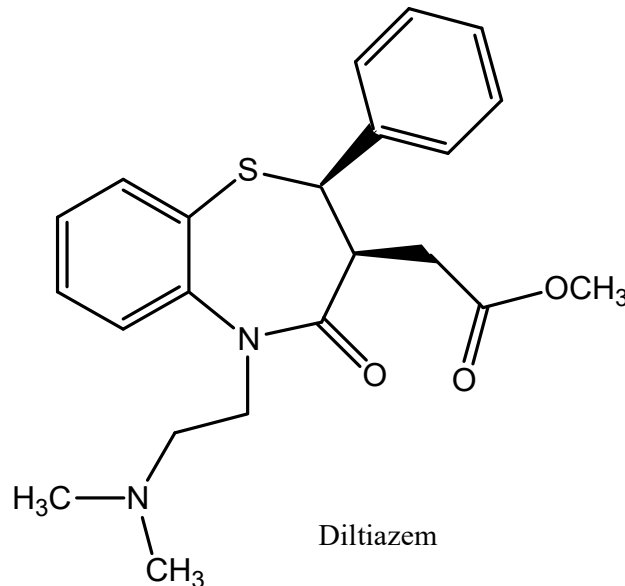
Isradipine



Nimodipine

Benzothiazepines

- Diltiazem
 - Vasodilating action is stereospecific for the *d-cis* isomer



Diltiazem

Antiarrhythmic drugs

- Cardiac arrhythmia is caused by
 - Disorders of impulse formation
 - Disturbance in conduction of nerve impulses through myocardial tissue
- Cardiac impulses generated by special tissues in the right atrium
 - Sinoatrial node
 - Or
 - Pacemaker cells

- Released impulse excites heart tissue in a pattern related to
 - Myocardial cell membrane potentials
 - Na & K fluxes in and out of the cells
- Disorders in impulse conduction can arise from
 - Blocking of some fibers
 - E.g. through localized myocardial disease (infarcts)

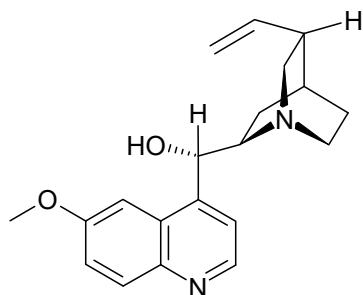
- Four classes of drugs are used in the treatment of arrhythmia:
 - Class-I: (sodium channels blockers).
 - Class-II (β -adrenergic blockers)
 - Class-III (potassium channels blockers)
 - Class-IV (cardioselective CCBs)

Class-I Antiarrhythmic Drugs

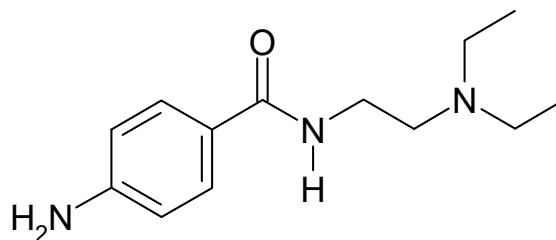
- They are called *membrane stabilizing* (*depressant*) drugs, act on sodium channels and block the depolarizing inward Na^+ current.
- They bind Na^+ channels in the *open* or *inactivated* state and dissociate from the channels in the *resting* state.
- They affect arrhythmic hearts more than normal hearts (use dependent channels blockade).

- According to the rate of dissociation from the sodium channels, *Class-I* drugs could be subdivided into:
 - *Class-IA* (intermediate rate of dissociation)
 - *Class-IB* (rapid rate of dissociation)
 - *Class-IC* (slow rate of dissociation)

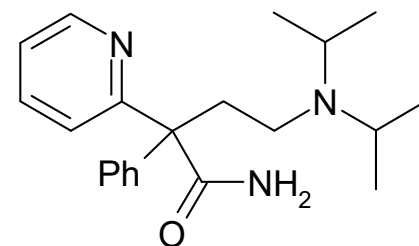
- **Class-IA Drugs**



Quinidine

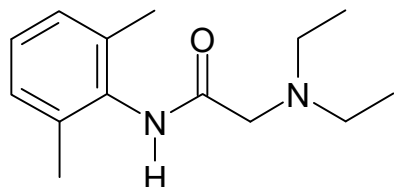


procainamide

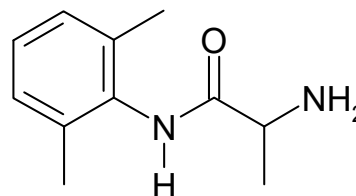


Disopyramide

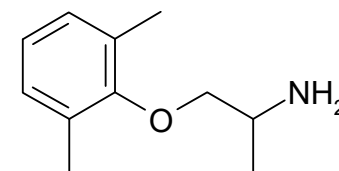
- **Class-IB Drugs**



Lidocaine

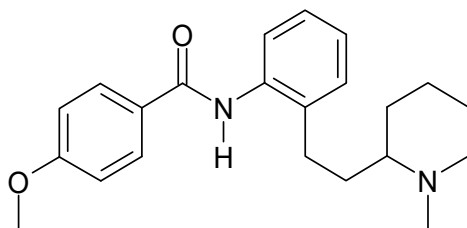


Cocainize



Mexiletine

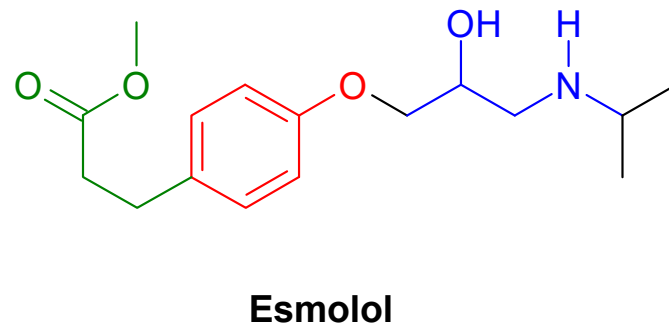
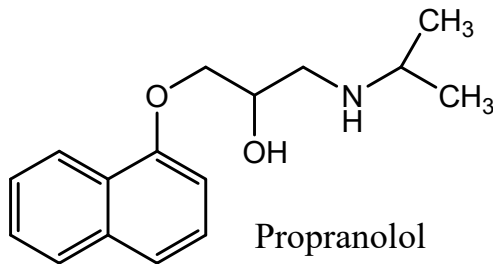
- **Class-IC Drugs**



Encainide

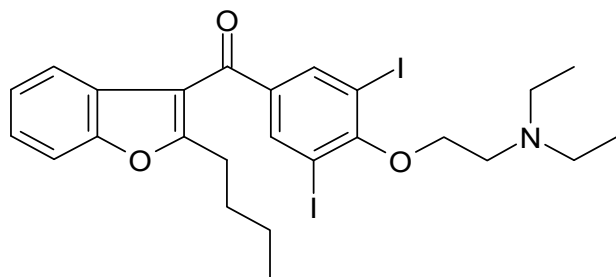
Class-II Antiarrhythmic Drugs

- They decrease inward calcium current by blocking the β -adrenergic receptors and inhibiting sympathetic activation of cardiac automaticity and conduction.
- Propranolol, Acebutolol, Metoprolol and Esmolol,

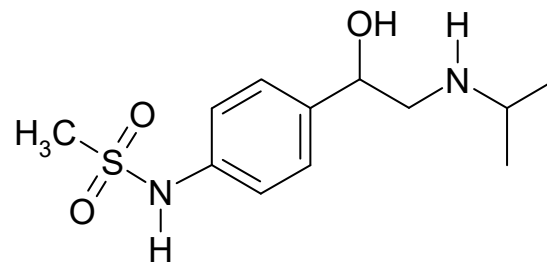


Class-III Antiarrhythmic Drugs

- Drugs from this class share the ability to block potassium channels; some members are able to block other channels as well.
- Potassium channels are activated during the repolarization (Phase 3) of the action potential; thus, their blockade prolongs action potential duration resulting in an increase in effective refractory period



Amiodarone



Sotalol

Class-IV Antiarrhythmic Drugs

- These are the cardio-selective CCBs namely Verapamil, Biperidil and Diltiazem
- They decrease the conduction velocity and increase the refractory period
- *Refer structures above*

Antihypertensive agents

- Hypertension = high blood pressure
- Usually defined as
 - Mild
 - Moderate
 - Severe
- Often referred to as the silent killer
 - Because patients feel well & have no obvious clinical symptoms

Hypertension ...

- Hypertension
 - Systolic BP 140 mm Hg or higher
 - Diastolic BP 90 mm Hg or higher
- 95 % of all cases are primary (essential) hypertension, the rest is either renal or due to endocrine abnormalities
- Risk of stroke or heart attack rises sharply when systolic pressure > 165mm Hg (normal 110-140)

Hypertension ...

- Remains major risk of stroke in patients bellow 65
 - Second only to diabetes as important cause of renal failure
- Hypertension often divided into two categories
 - Essential or primary hypertension
 - Secondary hypertension

- Primary hypertension

- Most common form

- Causes are complex & not fully understood

- » Enhanced adrenergic activity recognized as principal contributor

- If not treated, result in increased risk of death from

- » Stroke

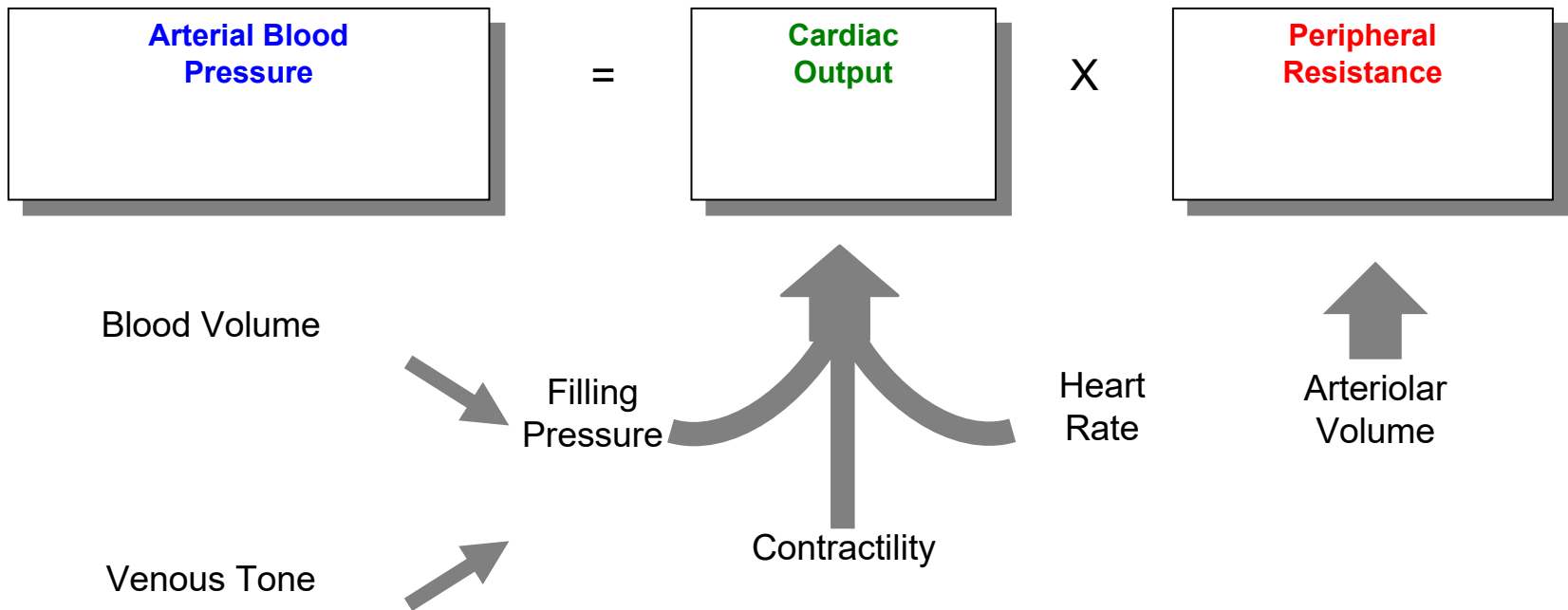
- » Heart attack

- » Heart failure

- » Kidney problems

- » Diabetes

- Secondary hypertension is a consequence of other diseases
 - That need their own treatments
 - Various kidney diseases
 - Hormonal disorders etc.



Classes of Antihypertensive Drugs

- Diuretics
- Sympatholytics
- Renin Angiotenin System (RAS) inhibitors
- Vasodilators

Diuretics:

- First line single drug treatment for mild hypertension
- Decrease BP by two mechanisms
 - Decrease Blood Volume so decrease COP
 - Decrease PVR by decreasing sodium content of the smooth muscles so decrease contractility

- Increase diuresis→ lowers blood volume
→ lower blood pressure
 - Thiazide diuretics (e.g. chlorothiazide) were introduced in the late 1950's
 - Were the first major antihypertensive drugs
 - Main criticism of diuretics
 - Cause hypokalemia → ↑ & possibly fatal cardiac arrhythmias

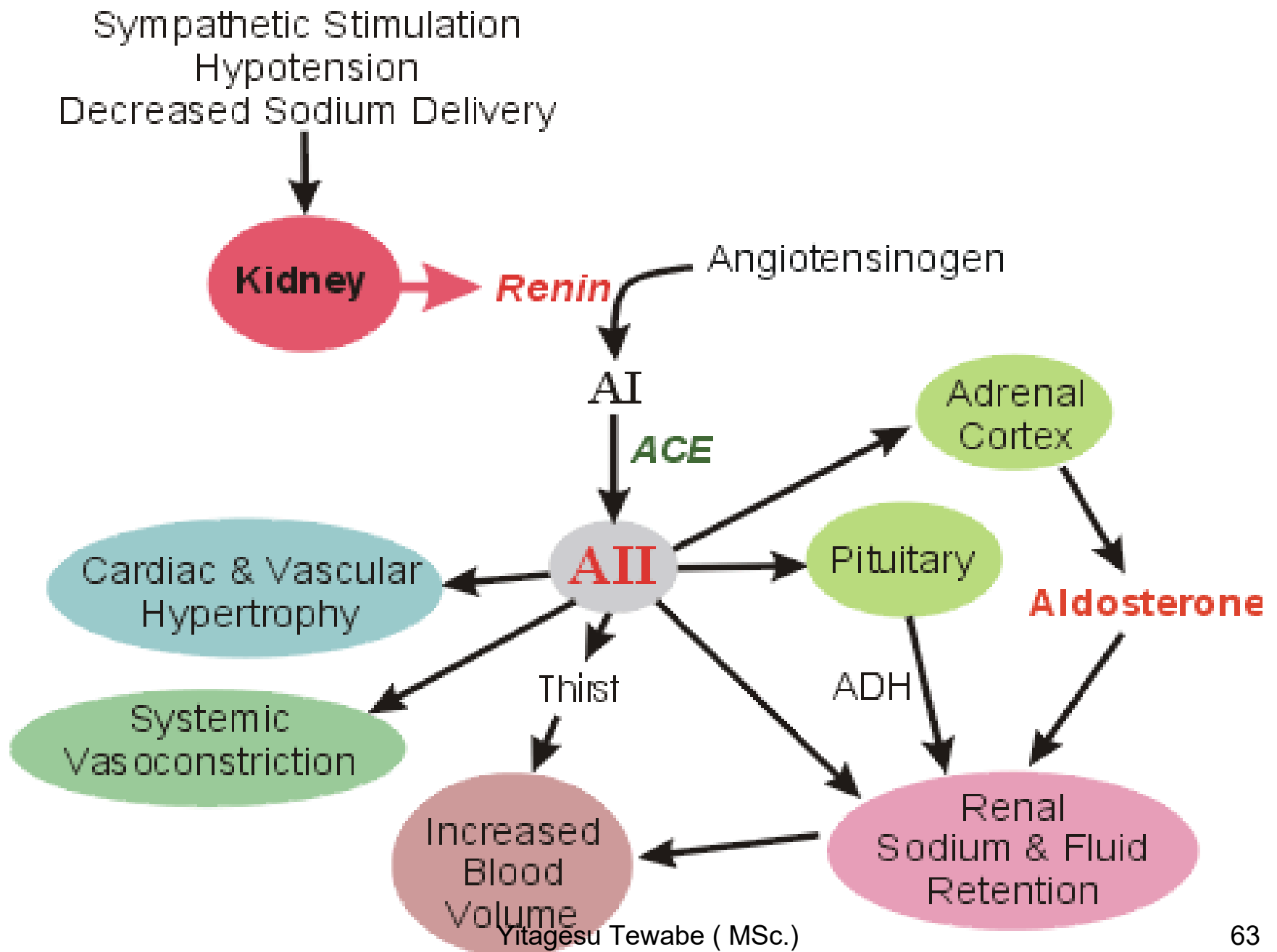
- **Sympatholytic:**
 - Adrenergic Receptor Blockers
 - Ganglionic Blockers
 - Agents depleting NE Stores
 - Centrally Acting Drugs

∇ **β-blockers**

- Act on β-adrenergic receptors
 - Reduce the rate at which the heart works → reduced blood output
 - Relax smooth muscle in arteries → reduction in peripheral resistance to blood flow

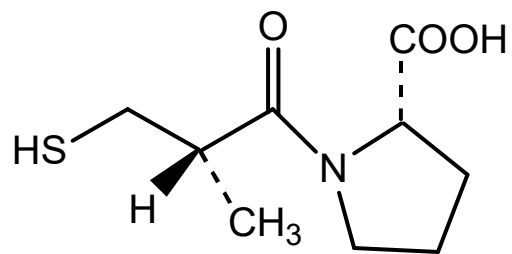
- Common adverse effects of β -blockers are
 - Easy fatigue with exercise
 - Frequent complaint of cold hands & feet
 - Due to reduced cardiac output
- Combination therapy of diuretics & β -blockers is very effective
 - Two drug types currently advised in the US for initiation of treatment

- **Calcium antagonists & ACE inhibitors**
 - Act through vasodilation
 - Enlargement of small blood vessels
 - Reduce resistance & hence pressure

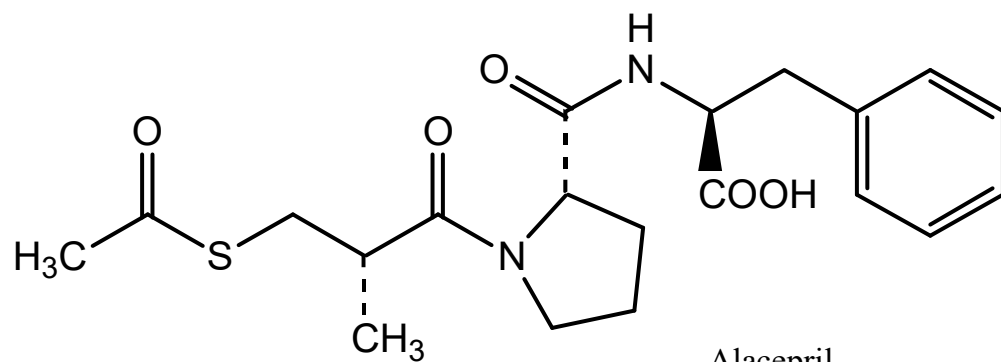


Angiotensin converting enzyme inhibitors (ACEI)

- **Captopril**
 - First orally active ACE inhibitor
 - ACE's active site contains Zn where the SH of captopril binds
 - Many retain significant structural features of captopril
 - One example is alacepril
 - However all lack specificity
 - Cleave peptides other than angiotensin I
 - Causing side effects

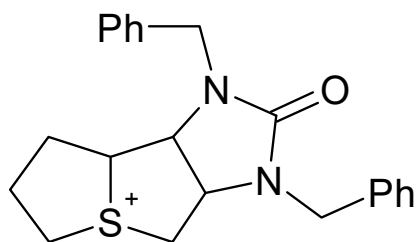


Captopril



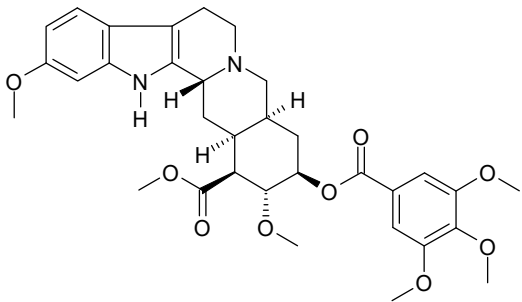
Alacepril

a. Ganglionic Blockers

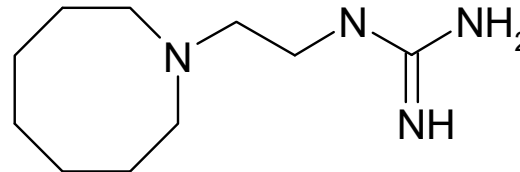


Trimethaphan

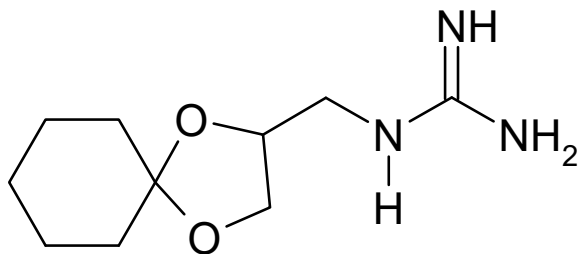
- **Agents Depleting NE Stores**



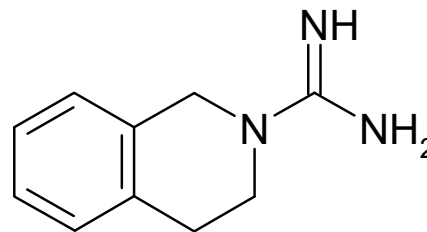
Reserpine



Guanethidine

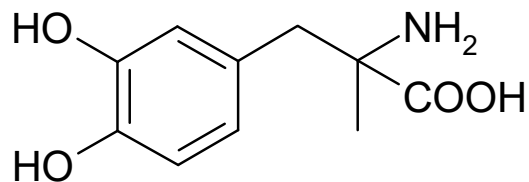


Guanadrel

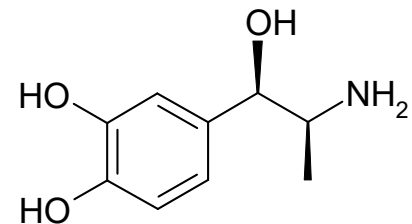


Debrisoquin

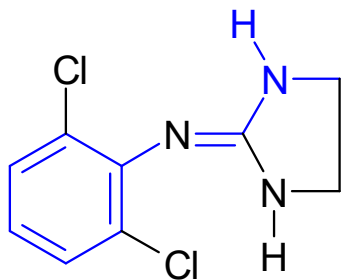
- **Centrally Acting Drugs:**



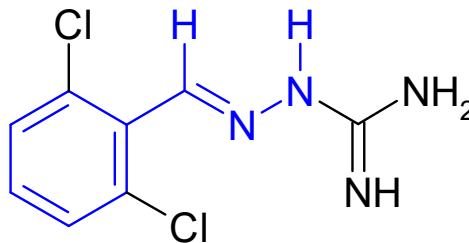
α -methyl dopa (*Aldomet*®)



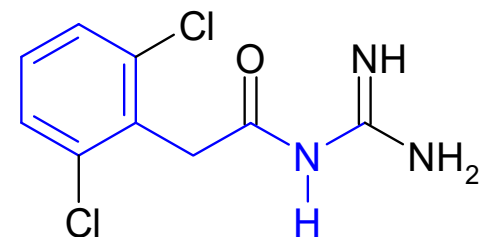
α -methyl NE



Clonidine



Guanabenz



Guanfacine